The foundations: six steps

From a raw recording to a short list of numbers per response. The foundations are in place; this is not yet a validated tool.

STEP 1

Read

Load the recording and its events.

STEP 2

Find the stimulations

Locate each pulse from the artifact it leaves.

STEP 3

Remove the artifact

Clean the few ms around each pulse.

STEP 4

Prepare

Baseline, noise level, reference, bad trials.

STEP 5

Test for a response

Is there a response, or only noise?

STEP 6

Describe the response

Fit bumps: latency, amplitude, width.

Steps 2 to 6 each hold settings that change the result a lot. The next slides list them, with the possible values and the value we start from.

STEP 2

Find the stimulations

Each pulse leaves a sharp jump in the signal. We score that jump against the noise and keep the jumps that are large and repeat.



Detection threshold

How many times the noise a jump must reach to count as a pulse.

z > 10 | z > 15 | **z > 20** | z > 30

Size gate

A candidate must reach this share of the size of the clear pulses.

none | 10 % | **25 %** | 50 %

Fallback score

If the median over channels finds nothing, use the channels that move most.

none | **90th percentile** | maximum

● starting value

STEP 3

Remove the artifact

Right after the pulse the signal is dominated by the stimulation itself. We mask that window and repair it.



[–] Masked window (CCEP)

Time around the electrical pulse that is masked.

shorter | **–2 to 8 ms** | longer

[–] Masked window (TMS)

Time around the magnetic pulse that is masked.

shorter | **–2 to 12 ms** | longer

Repair method

How the masked window is filled or the artifact subtracted.

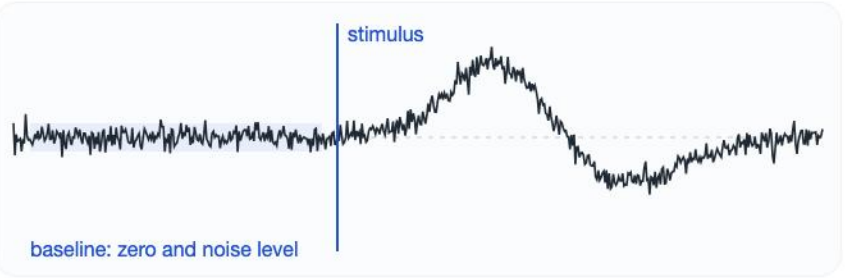
set to zero | interpolation | **template + tail (CCEP)** | **interpolation + subspace (TMS)** | ICA

● starting value

STEP 4

Prepare

Set the zero, measure the noise, choose the electrical reference and flag bad trials. The noise level drives every later test.



Baseline window

Quiet period before the stimulus, used for the zero and the noise.

shorter | **CCEP –500 to –20 ms** | **ERP –200 to 0 ms** | longer

Noise source

Where the noise level is measured.

baseline only | **auto: baseline or plus-minus**

Noise model

Noise remembers its past; correct for it or not.

none | effective n only | **AR whitening**

Electrical reference

What every electrode is measured against.

bipolar | **common average (TEP)** | **adaptive common average (CCEP)**

Trial rejection

Flag a trial if it is far from the others (z > 5), poor (quality < 0.5) or > 20 % masked; at most 30 % removed.

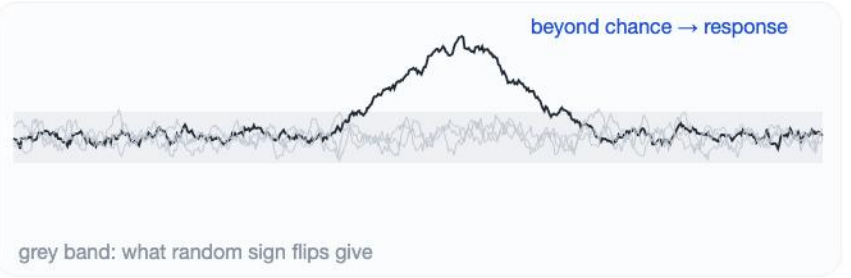
softer | **z 5 · 0.5 · 20 % · max 30 %** | stricter

● starting value

STEP 5

Test for a response

Before describing a response, check that it exists. We build chance by flipping the sign of trials at random and compare.



False-alarm risk α

Accepted chance of declaring a response that is only noise.

1 % · **5 %** · 10 %

Random sign flips

How many times chance is rebuilt; more gives a more precise p-value.

1 000 · 5 000 · 10 000

Many-electrode correction

Keeps false discoveries low when many electrodes are tested.

none · **FDR** · Bonferroni

Projection test

Null of the profile peak; more reliable with few trials.

half splits · disjoint pairs · **sign flips (≤ 4 096)**

Classic N1 rule (reference)

Published rule used only for comparison.

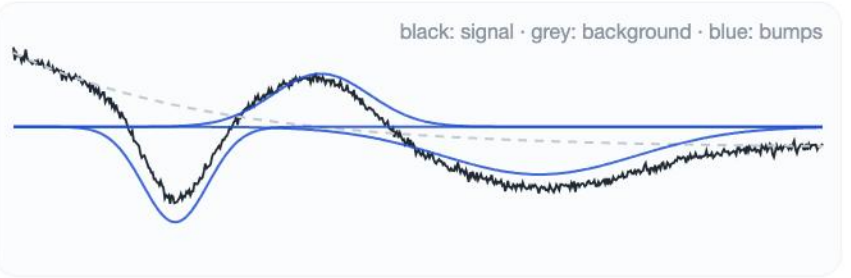
peak > 3.4 SD, 9–100 ms · other cut-offs

● starting value

STEP 6 · PART 1 — THE MODEL

Describe the response

The response is written as a background plus a few bumps. These settings decide which bumps the model may use and how many it keeps.



Bump shapes

Symmetric bell, skewed bell, or both.

Supported by tests

Gaussian | skewed | **both**

Background

Slow part under the bumps.

Supported by tests · tested on CCEP

none | **constant (ERP, TEP)** | linear | **exponential (CCEP)**

Width and number limits

Widest bump allowed; most bumps allowed.

Supported by tests · plateau

≤ 0.1 s | **0.25 s · 6 bumps** | 0.5 s · 10 bumps

Keep one more bump: ΔBIC

A bump is kept if the score drops by at least this much. The key setting.

Supported by tests · plateau 3–10

3 | 8 | **10** | 20 | calibrated

Starting candidate

Size a residual peak needs before a bump is tried (× noise).

Supported by tests · fastest without loss

1 σ | **2 σ** | 3 σ

Merge close bumps

Merge two bumps closer than c × ($\sigma_1 + \sigma_2$).

No measurable effect

0 | **0.5** | 2

Cancelling pairs

Reject a + and a – bump that cancel each other.

Supported by tests · ERP; no effect in CCEP

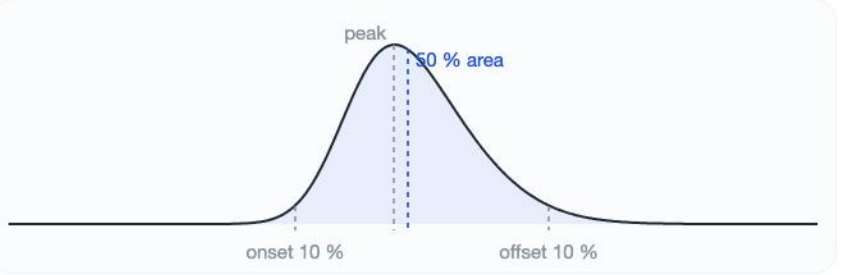
keep (CCEP) | **reject (ERP, TEP)**

● starting value Verdicts: version-2 tests (known truth, subject-level statistics, held-out half)

STEP 6 · PART 2 — THE MEASURES

Describe the response

Once the bumps are fitted, these settings decide how latency, onset and duration are read, and where components are looked for.



Latency definition

When a bump happens.

Supported by tests

raw peak | model peak | **50 % of the model area**

Onset and offset

First and last time the bump reaches this share of its height.

Questioned by tests · 20–50 % more accurate

5 % | **10 %** | 20 % | 50 %

Targeted ERP measure

Window margin, parts per component, tolerance for a match.

Questioned by tests · parts ok; ± 10 ms too strict

1 part | **3 parts** | **margin ± 25 ms** | **tolerance ± 10 ms** | tolerance ± 25 ms

TEP windows

Where N45 and N100 are looked for after a TMS pulse.

Inconclusive (too few subjects)

narrow | **N45 38–55 · N100 80–140 ms** | wide

Trial by trial

Largest shift a bump may take in one trial (× its width).

Questioned by tests · noise-dominated at any bound

0.1 | 0.25 (intracranial) | **0.5 (scalp)** | 1 | 2